

Part I

SUMMARY of the Limit Laws

Summary

February 21, 2026

0.1 The Limit Laws:

Continuity If a function f is **continuous at** a , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Composition Limit Law If a function f is **continuous at** $b = \lim_{x \rightarrow a} g(x)$, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

For the limit laws below, assume $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist. The following properties hold.

Sum Limit Law

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

Difference Limit Law

$$\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

Product Limit Law

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \left(\lim_{x \rightarrow a} f(x)\right) \cdot \left(\lim_{x \rightarrow a} g(x)\right)$$

Quotient Limit Law

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \text{ provided } \lim_{x \rightarrow a} g(x) \neq 0$$

Limit laws also hold for one-sided limits.

Part II

Recitation Questions

Problem 1

February 21, 2026

Problem 1 The following argument shows

$$\lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} = \frac{135 - 4\sqrt{3}}{\sqrt{156}}.$$

State which limit law is used to justify each step. (A particular step may have more than one limit law as a justification.)

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} &= \frac{\lim_{x \rightarrow 3} (5x^3 - 4\sqrt{x})}{\lim_{x \rightarrow 3} \sqrt{x^5 - 87}} \\ &= \frac{\left(\lim_{x \rightarrow 3} 5 \right) \left(\lim_{x \rightarrow 3} (x^3) \right) - \left(\lim_{x \rightarrow 3} 4 \right) \left(\lim_{x \rightarrow 3} \sqrt{x} \right)}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} \\ &= \frac{5 \lim_{x \rightarrow 3} (x^3) - 4 \lim_{x \rightarrow 3} \sqrt{x}}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} \\ &= \frac{5(\lim_{x \rightarrow 3} x)^3 - 4\sqrt{3}}{\sqrt{\lim_{x \rightarrow 3} (x^5) - \lim_{x \rightarrow 3} (87)}} \\ &= \frac{5(3)^3 - 4\sqrt{3}}{\sqrt{3^5 - 87}} \\ &= \frac{135 - 4\sqrt{3}}{\sqrt{156}} \end{aligned}$$

Explanation.

$$\begin{aligned}
 \lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} &= \frac{\lim_{x \rightarrow 3} (5x^3 - 4\sqrt{x})}{\lim_{x \rightarrow 3} \sqrt{x^5 - 87}} && \text{Quotient law} \\
 &= \frac{\left(\lim_{x \rightarrow 3} 5 \right) \left(\lim_{x \rightarrow 3} (x^3) \right) - \left(\lim_{x \rightarrow 3} 4 \right) \left(\lim_{x \rightarrow 3} \sqrt{x} \right)}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} && \text{Product law, Composition} \\
 &= \frac{5 \lim_{x \rightarrow 3} (x^3) - 4 \lim_{x \rightarrow 3} \sqrt{x}}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} && \text{Continuity} \\
 &= \frac{5(\lim_{x \rightarrow 3} x)^3 - 4\sqrt{3}}{\sqrt{\lim_{x \rightarrow 3} (x^5) - \lim_{x \rightarrow 3} (87)}} && \text{Difference, Continuity} \\
 &= \frac{5(3)^3 - 4\sqrt{3}}{\sqrt{3^5 - 87}} && \text{Continuity} \\
 &= \frac{135 - 4\sqrt{3}}{\sqrt{156}}
 \end{aligned}$$

Problem 2

February 21, 2026

Problem 2 Find the limit and justify your answer.

(a) $\lim_{x \rightarrow 0} |x|$

Explanation. Remember $|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$
 We first compute the one-sided limit: $\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$.

Next, we compute the other one-sided limit:

$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = \left(\lim_{x \rightarrow 0^-} (-1) \right) \left(\lim_{x \rightarrow 0^-} x \right) = (-1)(0) = 0.$$

Therefore $\lim_{x \rightarrow 0^-} |x| = 0 = \lim_{x \rightarrow 0^+} |x|$. Hence, $\lim_{x \rightarrow 0} |x| = 0$.

(b) $\lim_{x \rightarrow 2} \ln \left(\sin(x-2) + e^x \sin \left(\frac{\pi x}{4} \right) \right)$

Explanation.

$$\begin{aligned} \lim_{x \rightarrow 2} \ln \left(\sin(x-2) + e^x \sin \left(\frac{\pi x}{4} \right) \right) &= \ln \left(\lim_{x \rightarrow 2} \left(\sin(x-2) + e^x \sin \left(\frac{\pi x}{4} \right) \right) \right) \\ &= \ln \left(\lim_{x \rightarrow 2} \sin(x-2) + \lim_{x \rightarrow 2} \left(e^x \sin \left(\frac{\pi x}{4} \right) \right) \right) \\ &= \ln \left(\sin \left(\lim_{x \rightarrow 2} (x-2) \right) + \left(\lim_{x \rightarrow 2} e^x \right) \left(\lim_{x \rightarrow 2} \sin \left(\frac{\pi x}{4} \right) \right) \right) \\ &= \ln \left(\sin(0) + \left(\lim_{x \rightarrow 2} e^x \right) \sin \left(\lim_{x \rightarrow 2} \frac{\pi x}{4} \right) \right) \\ &= \ln \left(0 + e^2 \sin \left(\frac{2\pi}{4} \right) \right) \\ &= \ln \left(e^2 \sin \left(\frac{\pi}{2} \right) \right) = \ln(e^2(1)) \\ &= \ln(e^2) = 2 \end{aligned}$$

(c) $\lim_{x \rightarrow 1} \frac{3 + 2 \cos \left(\frac{\pi x}{3} \right)}{4x - 2x^3}$

Explanation.

$$\begin{aligned}
 \lim_{x \rightarrow 1} \frac{3 + 2 \cos\left(\frac{\pi x}{3}\right)}{4x - 2x^3} &= \frac{\lim_{x \rightarrow 1} \left(3 + 2 \cos\left(\frac{\pi x}{3}\right)\right)}{\lim_{x \rightarrow 1} (4x - 2x^3)} \\
 &= \frac{\lim_{x \rightarrow 1} (3) + \lim_{x \rightarrow 1} \left(2 \cos\left(\frac{\pi x}{3}\right)\right)}{4(1) - 2(1)^3} \\
 &= \frac{3 + \left(\lim_{x \rightarrow 1} 2\right) \left(\lim_{x \rightarrow 1} \cos\left(\frac{\pi x}{3}\right)\right)}{4 - 2(1)} \\
 &= \frac{3 + 2 \cos\left(\lim_{x \rightarrow 1} \frac{\pi x}{3}\right)}{4 - 2} \\
 &= \frac{3 + 2 \cos\left(\frac{\pi}{3}\right)}{2} = \frac{3 + 2\left(\frac{1}{2}\right)}{2} \\
 &= \frac{3 + 1}{2} = 2.
 \end{aligned}$$

Problem 3

February 21, 2026

Problem 3 Suppose $f(x) = \begin{cases} x^2 - ax & \text{if } x < 3 \\ a2^x + 7 + a & \text{if } x > 3 \end{cases}$

Find a so that $\lim_{x \rightarrow 3} f(x)$ exists.

Explanation. Remember, it's not enough to find a value. We must also justify that the limit exists for the value chosen.

To find a number a for which $\lim_{x \rightarrow 3} f(x)$ exists we find a such that $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$.

To find the left-sided limit:

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2 - ax) = 9 - 3a.$$

To find the right-sided limit:

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (a2^x + 7 + a) = a2^3 + 7 + a = 9a + 7.$$

In order for $\lim_{x \rightarrow 3} f(x)$ to exist we must have the two one-sided limits equal. That is,

$$\begin{aligned} \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^+} f(x) \\ 9 - 3a &= 9a + 7 \\ 2 &= 12a \\ a &= 1/6. \end{aligned}$$

Problem 4

February 21, 2026

Problem 4 Determine the value of $\lim_{x \rightarrow 0} \left(x^2 \cos \left(\frac{1}{x} \right) \right)$. **EXPLAIN.**

Explanation. $\lim_{x \rightarrow 0} \cos \left(\frac{1}{x} \right)$ does not exist, so we cannot apply the Product law.

We know that for any $x \neq 0$ that $-1 \leq \cos \left(\frac{1}{x} \right) \leq 1$. Multiplying this inequality by x^2 (which is positive) we find that for all x near 0,

$$-x^2 \leq x^2 \cdot \cos \left(\frac{1}{x} \right) \leq x^2$$

The limits of the functions on the left and right sides of this compound inequality are:

$$\begin{aligned} \lim_{x \rightarrow 0} (-x^2) &= 0 \\ \lim_{x \rightarrow 0} x^2 &= 0 \end{aligned}$$

Our explanation:

$\lim_{x \rightarrow 0} \left(x^2 \cos \left(\frac{1}{x} \right) \right) = 0$ by the Squeeze Theorem since $-x^2 \leq x^2 \cdot \cos \left(\frac{1}{x} \right) \leq x^2$ and $\lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$.

Problem 5

February 21, 2026

Problem 5 For all x near 0, the inequalities $1 - \frac{x^2}{6} \leq \frac{\sin(x)}{x} \leq 1$ are true.

Use these inequalities to find $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$. **EXPLAIN.**

Explanation. $1 - x^2/6$ is a polynomial, so it is continuous at $x = 0$. That means $\lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6}\right) = 1$. The function 1 is continuous since it is a constant, so $\lim_{x \rightarrow 0} 1 = 1$.

That means $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$ by the Squeeze Theorem, since $1 - \frac{x^2}{6} \leq \frac{\sin(x)}{x} \leq 1$ and $\lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6}\right) = \lim_{x \rightarrow 0} 1 = 1$.

Problem 6

February 21, 2026

Problem 6

Problem 6 Two functions, h and g , are given

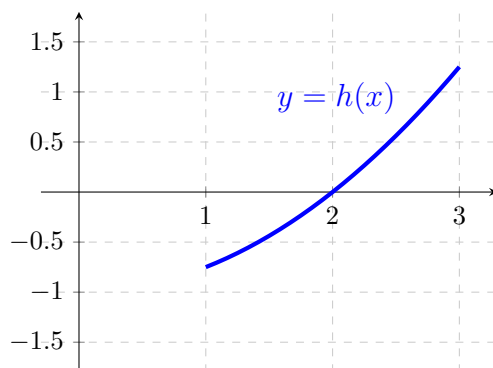
$$h(x) = \frac{x^2 - 4}{4}, \quad 1 < x < 3$$

$$g(x) = x - 2, \quad 1 < x < 3$$

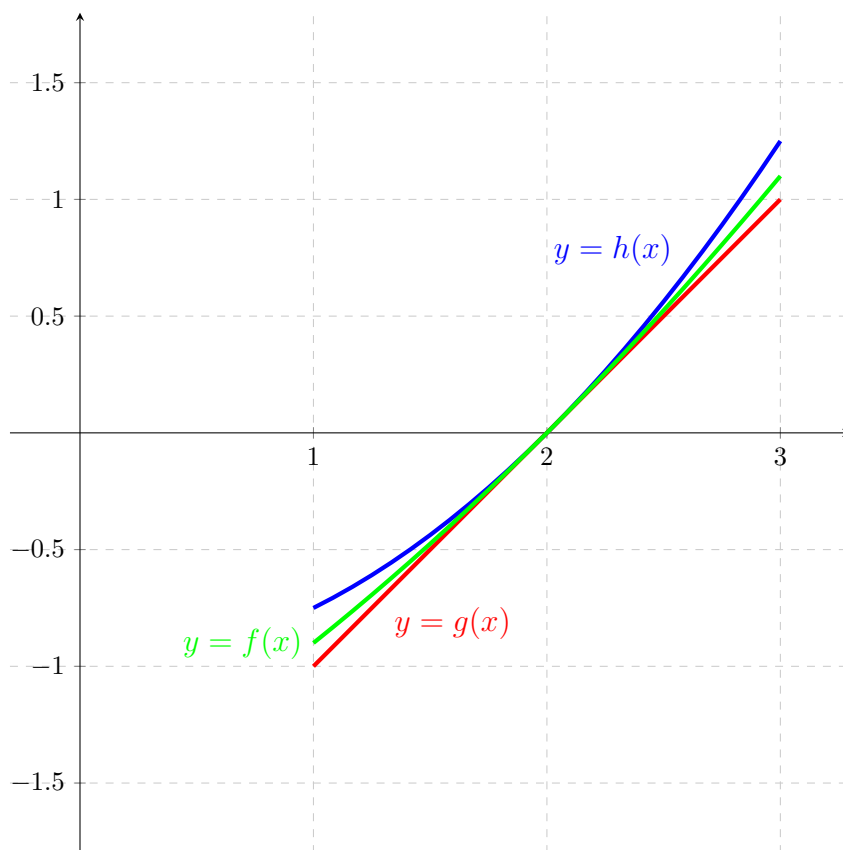
The graph of the function h is given in the figure below. Let f be a function defined on the interval $(1, 3)$ that satisfies the following inequalities

$$g(x) \leq f(x) \leq h(x), \quad 1 < x < 3$$

- (a) In the figure below, sketch and label the graph of g and a possible graph of f . (All three functions have a common domain $(1, 3)$.)



Explanation.



(b) Evaluate the limit, or state that it does not exist. Justify your answer.

(i) $\lim_{x \rightarrow 2} f(x)$

Explanation. g and h are both continuous function since they are both polynomials so we have:

$$\lim_{x \rightarrow 2} g(x) = g(2) = 0$$

$$\lim_{x \rightarrow 2} h(x) = h(2) = 0$$

We have $g(x) \leq f(x) \leq h(x)$ so by the squeeze theorem, $\lim_{x \rightarrow 2} f(x) = 0$ since $g(x) \leq f(x) \leq h(x)$ with $\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} h(x) = 0$.

(ii) $\lim_{x \rightarrow 2} \frac{f(x) + 2}{x - 1}$

Explanation.

$$\lim_{x \rightarrow 2} \frac{f(x) + 2}{x - 1} = \frac{\lim_{x \rightarrow 2} (f(x) + 2)}{\lim_{x \rightarrow 2} (x - 1)} = \frac{0 + 2}{2 - 1} = \frac{2}{1} = 2$$

Problem 6

(iii) $\lim_{x \rightarrow 2} g(1 + e^{f(x)})$

Explanation.

$$\begin{aligned}\lim_{x \rightarrow 2} g(1 + e^{f(x)}) &= g\left(\lim_{x \rightarrow 2} (1 + e^{f(x)})\right) = g\left(\lim_{x \rightarrow 2} 1 + \lim_{x \rightarrow 2} e^{f(x)}\right) = g\left(1 + \lim_{x \rightarrow 2} e^{f(x)}\right) \\ &= g(1 + e^0) = g(2) = 0\end{aligned}$$